

QUANTUM CONSTRAINT FRAMEWORK AT SPACETIME SINGULARITIES

A Postulate of Isomorphic Cosmic Equilibrium

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ABSTRACT

General Relativity (GR) provides a robust geometric description of spacetime curvature ($G_{\mu\nu}$), culminating in the classical definition of the Schwarzschild event horizon ($r = \frac{2GM}{c^2}$). At this boundary, standard GR predicts a catastrophic collapse into a gravitational singularity of infinite density—a result fundamentally forbidden by the unitary bounds of Quantum Mechanics (QM). The mathematical intersection between these two domains remains unresolved.

This paper introduces the **Quantum Constraint Framework (QCF)**, which posits that the convergence at the event horizon is not a mathematical failure of the smooth classical continuum, but rather a manifestation of a fundamental macro-to-micro structural equilibrium.

I. THE CORE POSTULATE

We formally propose the foundational axiom governing boundary convergence:

$$\sum \left(r = \frac{2GM}{c^2} \right) = \sum QM$$

This equation asserts that the invariant physical quantity expressed via macroscopic spacetime geometry at the horizon is structurally isomorphic to the total quantum state space ($\sum QM$). Under the QCF, the event horizon does not function merely as a classical point of no return, but as a dynamic thermodynamic interface where gravitational

degrees of freedom are bounded and regulated by quantum mechanical conservation laws. The universe requires two opposing, localized languages to function:

- 1. The Macro Continuum ($\sum$ GR):** Binds the deterministic, large-scale structural scaffolding of the cosmos.
- 2. The Micro Probabilistic ($\sum$ QM):** Binds the discrete, chaotic kinetic energy states.

II. IMPLICATIONS FOR PHYSICAL SYSTEMS

The framework provides a direct, non-perturbative pathway toward resolving the Black Hole Information Paradox. By enforcing this balance equation, the absolute loss of quantum information to an infinite singularity is geometrically strictly forbidden. Instead, the horizon acts as a transactional boundary where classical curvature is continuously modulated by the quantum sum, effectively truncating infinities.

Furthermore, apparent experimental deviations from standard model predictions (e.g., localized particle anomalies) are reframed not as systemic refutations of equilibrium, but as critical, quantifiable signposts indicating unmeasured or latent dark sectors within $\sum$ QM.

III. CONCLUSION & METHODOLOGY

The Quantum Constraint Framework shifts the paradigm from a forced mathematical unification to a boundary-driven dualistic isomorphism. The equilibrium condition defines a strict, non-negotiable selection criterion to which any viable theory of Quantum Gravity must adhere. Future work will focus on explicitly defining the functional mathematical operator for $\sum$ QM capable of mapping onto the localized GR metric tensor.

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